

Math Primer

(14. Numerical Integration)

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Thomas Simpson
(1710-1761)

Numerical Integration – Motivation

- ▶ Often, we want to evaluate an integral for which no analytical expression exists. In this case, we resort to numerical techniques where we discretize the domain into smaller sub-intervals and perform a summation of the smaller areas.
- ▶ The need for integral is of paramount importance in Quant Finance. Some basic applications are given below

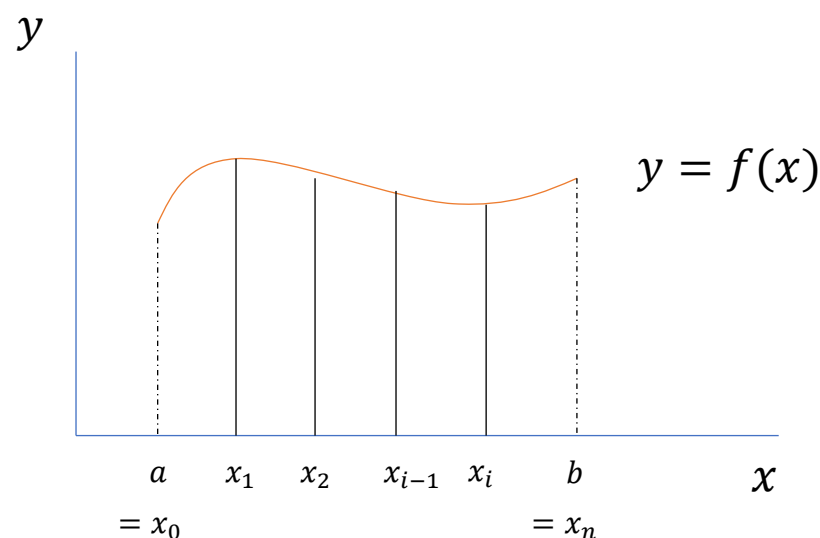
$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \quad (\text{CDF of Normal distribution})$$

$$\mathbb{E}[h(X)] = \int_{-\infty}^{\infty} h(x) f_X(x) dx \quad (\text{Price of a derivative as Expectation})$$

The Basic Set up



► Let's consider the following integral $\int_a^b f(x)dx$

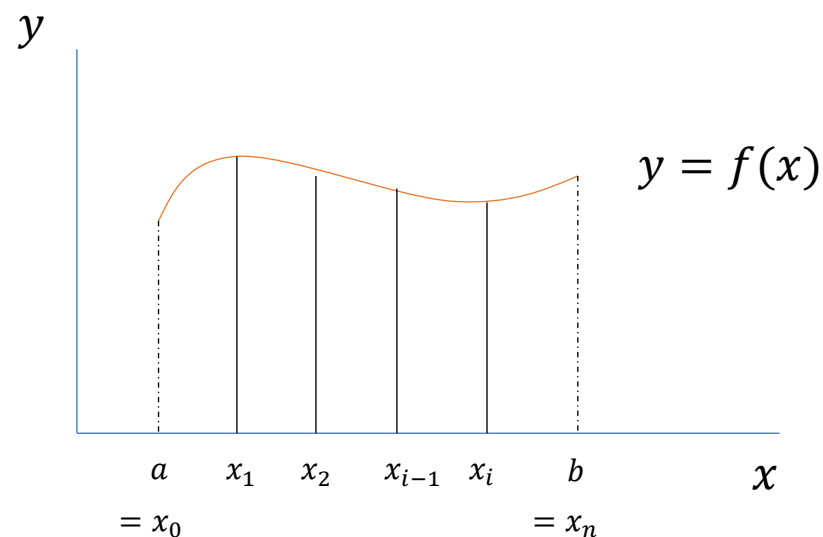


- We divide the interval $[a, b]$ into n sub-intervals : $a = x_0 < x_1 < x_2 < \dots < x_n = b$.
- Calculate the approximate area of under the curve $y = f(x)$ for each of the n sub-intervals
- Aggregate.

The Basic Set up



► Let's write down some basic equations



(Terminal Points)

$$x_0 = a$$
$$x_n = b$$

Sub-interval Width (h)

$$h = \frac{b - a}{n}$$

Intermediate Points (x_i)

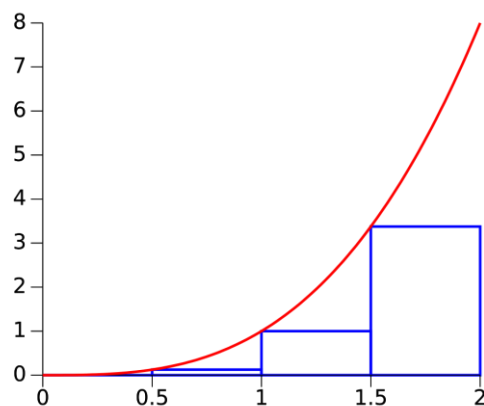
$$x_i = x_0 + ih$$

1. Riemann Sum

- Riemann sum approximates the integral by treating the partitions as rectangles.

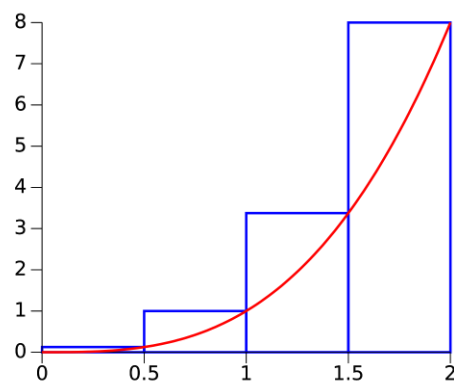
$$S = \sum_{i=1}^n f(x^*)h$$

- We have the following three variants of Riemann Sum.



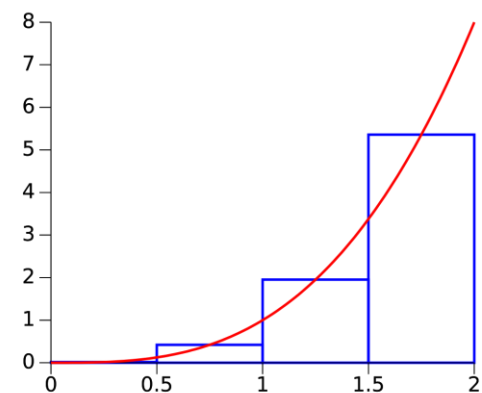
(Left Rule) $\sim \mathcal{O}(h^2)$

$$x^* = x_{i-1}$$



(Right Rule) $\sim \mathcal{O}(h^2)$

$$x^* = x_i$$

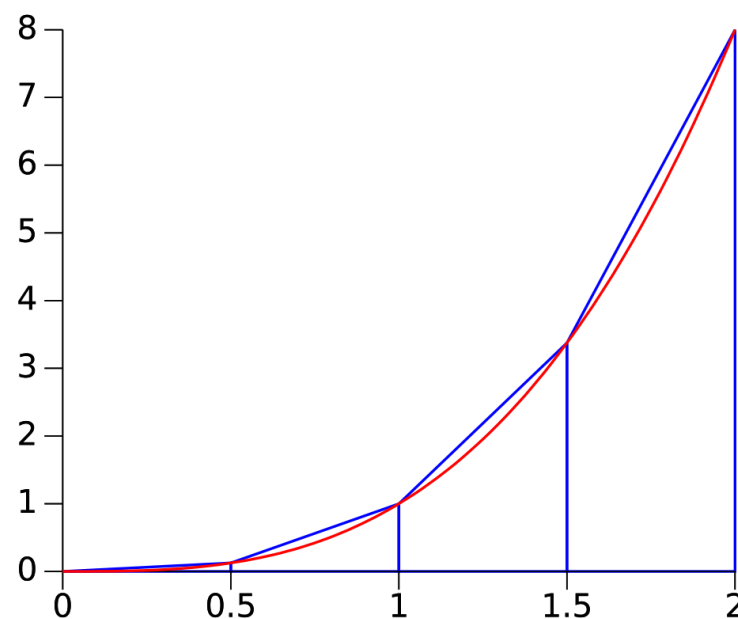


(Middle Rule) $\sim \mathcal{O}(h^3)$

$$x^* = (x_{i-1} + x_i)/2$$

2. Trapezoidal Rule

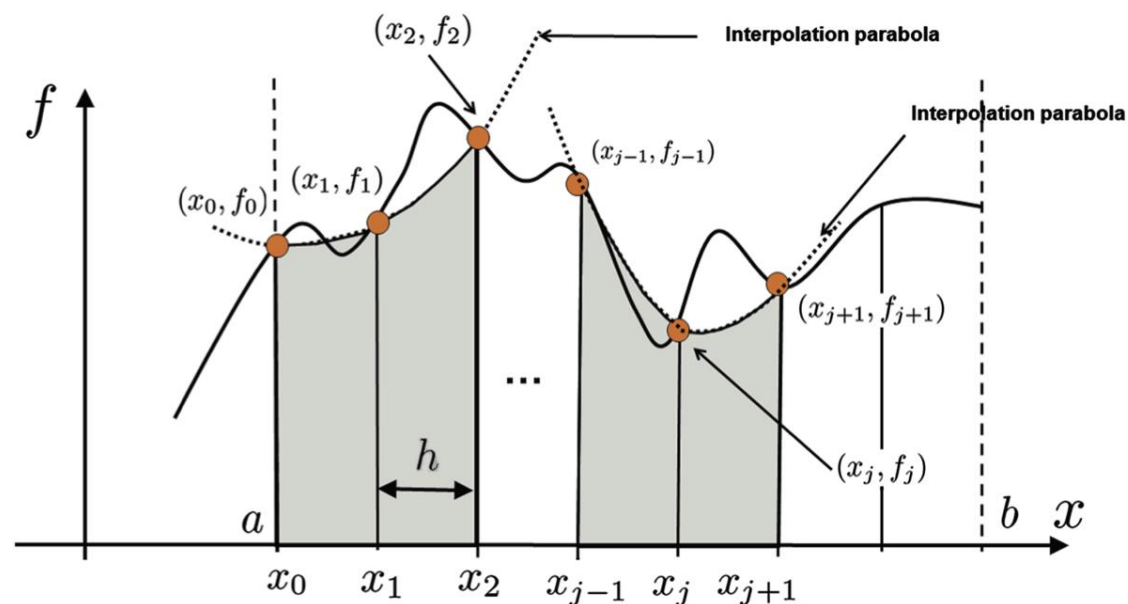
- Trapezoidal Rule approximates the strips as trapeziums. Note that it's not the same as mid-point rule. Mid-point rule evaluates the function at the middle point of x , whereas trapezoidal rule takes the average of the function at left and right values



The area is given as $S = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \cdots + y_{n-1})]$, order of convergence $\mathcal{O}(h^3)$

3. Simpson's 1/3rd Rule

- We saw that the trapezoidal rule interpolates the function points linearly. Simpson's 1/3rd rule on the other hand interpolates using a quadratic function (parabola). Of course, it needs 3 such points to fit a parabola $y = ax^2 + bx + c$



- To use Simpson's 1/3rd rule, we must have **n as an even number**.
- The area is given as
$$S = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \cdots + y_{n-1}) + 2(y_2 + y_4 + \cdots + y_n)]$$

4. Simpson's $3/8^{\text{th}}$ Rule



- ▶ Simpson's $3/8^{\text{th}}$ rule interpolates using a cubic function between 4 consecutive points.
- ▶ To use Simpson's $3/8^{\text{th}}$ rule, we must have **n as a multiple of 3**.
- ▶ The area is given as
$$S = \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_2 + \cdots + y_{n-1}) + 2(y_3 + y_6 + \cdots + y_n)]$$

5. Gaussian Quadrature

► All the formulae, we have seen so far employ a form

$$S = \sum_{i=0}^n w_i f(x_i)$$

We have dealt with uniform sub-intervals till now. What if we can choose our sample points more efficiently with flexible spacing ? Gaussian quadrature is one such scheme which allows us choose points more intelligently.

Before we proceed with the scheme, let's focus on the following integral. Note that any integral can be transformed to the following integral using variable transformation.

$$\int_a^b f(x) dx = \left(\frac{b-a}{2}\right) \int_{-1}^1 f\left(\left(\frac{b-a}{2}\right)t + \left(\frac{a+b}{2}\right)\right) dt \quad \text{by substituting} \quad x = \left(\frac{b-a}{2}\right)t + \left(\frac{a+b}{2}\right)$$



5. Gaussian Quadrature

► **Linear Case** : We can exactly match the integral of a linear polynomial by choosing the right w and the right t_0 using the following specification

$$\int_{-1}^1 f(t) dt = w_0 f(t_0)$$

If $f(t) = a_0 + a_1 t$, then it's easy to see that $w_0 = 2$ and $t_0 = 0$

This means that for a linear polynomial $f(t)$,

$$\int_{-1}^1 f(t) dt = 2f(0)$$

Verify for $f(t) = 2t + 4$



5. Gaussian Quadrature

► **Cubic Case** : We can exactly match the integral of a cubic polynomial by choosing the 4 parameters $\langle w_1, t_1, w_2, t_2 \rangle$ using the following specification

$$\int_{-1}^1 f(t) dt = w_1 f(t_1) + w_2 f(t_2)$$

If $f(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$, then we can show that

$$w_1 = w_2 = 1 \text{ \&}$$

$$t_1 = \frac{1}{\sqrt{3}} \quad t_2 = -\frac{1}{\sqrt{3}}$$

So up to a Cubic polynomial $\int_{-1}^1 f(t) dt = f\left(\frac{1}{\sqrt{3}}\right) + f\left(-\frac{1}{\sqrt{3}}\right)$



5. Gaussian Quadrature

- ▶ These calculations have been performed till much higher order polynomial (order 127) and exists as codified knowledge which can be simply looked up and used.
- ▶ Any function can be approximated as a high degree polynomial function (Taylor Series) and its integral can be performed using Gaussian Quadrature to remarkable accuracy.

As an example, if $f(x) = \frac{1}{x}$, then $\int_1^2 f(x) dx = \frac{1}{2} \int_{-1}^1 f\left(\frac{t+3}{2}\right) dt$

This integral is evaluated with lot of precision using only 5 sample points

i	t	$f\left(\frac{t+3}{2}\right)$	w
1	0	0.666666667	0.568889
2	-0.538469	0.812502565	0.478629
3	0.538469	0.56521615	0.478629
4	-0.90618	0.955191876	0.236927
5	0.90618	0.512009195	0.236927

True Value : 0.693147180559945

GQ(Approx.) : 0.693147157853039